

Predictive Stability and Adaptation Under Distribution Shift: Calibration, Discrimination, and Decision Thresholds in Credit-Risk Modeling

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Abstract

Probability models are commonly validated on held-out observations drawn from the same underlying environment as the development data, yet deployment populations may differ in ways that affect probability accuracy, calibration, discrimination, and decision rules differently. This study examines those distinctions using an empirical loan-default application and controlled Monte Carlo experiments. The empirical analysis uses 255,347 loan records with 29,653 recorded defaults (11.6%) to establish an internally validated probability model. Raw logistic regression achieved the lowest validation log loss among the candidate models and obtained test ROC-AUC 0.7585, PR-AUC 0.3288, Brier score 0.0908, and log loss 0.3111.

Controlled experiments then isolate three forms of distributional change: covariate shift combined with model misspecification, an intercept change in the conditional outcome mechanism, and a change in a predictor effect. Across 500 Monte Carlo replications per condition, these mechanisms produce qualitatively different failure patterns. A second set of experiments evaluates adaptation through intercept recalibration, intercept-and-slope recalibration, importance-weighted refitting where applicable, deployment refitting, and decision-threshold revision. Under intercept shift, simple recalibration nearly restores probability performance. Under covariate shift with misspecification, recalibration improves performance but refitting provides greater repair. Under severe predictor-effect shift, score recalibration improves some probability metrics but cannot reconstruct the changed multivariable relationship; deployment refitting restores discrimination substantially more effectively. Threshold revision can materially improve decision performance even when it does not repair the probability model itself.

The results support a failure-response view of model maintenance: the appropriate intervention depends on which component of the predictive system has become unstable. Stable discrimination does not guarantee stable probabilities, and recalibrating probabilities does not necessarily restore ranking or decision performance.

Keywords: distribution shift; probability calibration; discrimination; decision thresholds; model adaptation; credit risk; Monte Carlo simulation

1 Introduction

A probability model can perform well during internal validation and still behave differently after deployment. Dataset shift is the general setting in which the joint distribution of predictors and outcomes

differs between development and use (Quiñonero-Candela et al. 2009). The practical difficulty is not merely that aggregate performance may deteriorate. Different properties of the predictive system may change in different ways: ranking may remain useful while absolute probabilities become biased, or probability estimates may remain informative while a threshold selected in the development population becomes inappropriate.

Let $P(X, Y)$ denote the development distribution and $Q(X, Y)$ the deployment distribution. The factorization

$$P(X, Y) = P(Y | X)P(X)$$

separates changes in the predictor population from changes in the conditional outcome mechanism. This distinction is central to the present study because the statistical consequences of a change in $P(X)$ need not resemble those of a change in $P(Y | X)$.

Credit-risk modeling provides a natural application. Predicted probabilities of default are used to support risk assessment and threshold-based decisions, while the underlying borrower population, economic environment, and predictor–outcome relationships can evolve over time. Prior credit-scoring research has documented the practical importance of population and concept drift and has proposed dynamic or locally adaptive modeling approaches (Sousa et al. 2016; Nikolaidis and Doumpos 2022). Recent work has continued this direction by explicitly studying discrimination and calibration of credit-risk models under multiple forms of data drift (Peng and Lessmann 2026).

The empirical loan data in this study are used to establish an internally validated baseline rather than to claim observed deployment drift. The subsequent Monte Carlo experiments deliberately perturb a known data-generating process so that the consequences of covariate shift, baseline-risk shift, and predictor-effect shift can be isolated. A second simulation layer evaluates interventions that act on different components of the predictive system.

The study makes three contributions. First, it compares how distinct shift mechanisms affect probability accuracy, calibration, discrimination, and decision-threshold stability. Second, it evaluates recalibration, importance-weighted refitting, deployment refitting, and threshold revision under the same controlled conditions. Third, it links the resulting failure patterns to a mechanism-based response framework that distinguishes probability-model repair, ranking repair, and decision-rule repair.

The central question is therefore:

Which intervention repairs which type of predictive failure?

2 Literature Review

2.1 Distribution shift and predictive stability

The general problem of dataset shift arises when the distribution generating deployment observations differs from the distribution used for model development (Quiñonero-Candela et al. 2009). A useful decomposition distinguishes changes in the covariate distribution from changes in the conditional outcome mechanism. Under covariate shift, the marginal distribution of predictors changes while the conditional relationship $P(Y | X)$ remains stable. Shimodaira (2000) showed that density-ratio weighting can target predictive inference toward a shifted covariate distribution, with particular relevance when the fitted model is misspecified.

This distinction is important because observed predictive performance depends jointly on the conditional model and the population in which it is evaluated. Changes in case mix can alter discrimination even when the conditional relationship remains valid (Debray et al. 2015). Consequently, a change in ROC-AUC is not by itself sufficient evidence that the underlying probability model has become structurally incorrect. Modern guidance on prediction-model evaluation likewise emphasizes assessment in representative target populations and separation of development performance from performance in new settings (Collins et al. 2024).

2.2 Calibration, discrimination, and model updating

Calibration and discrimination describe different properties of a probability model. Discrimination concerns ordering: whether observations with events tend to receive higher scores than observations without events. Calibration concerns the numerical agreement between predicted probabilities and observed outcome frequencies (Steyerberg et al. 2010; Van Calster et al. 2019). A model can therefore retain useful discrimination while systematically underpredicting or overpredicting absolute risk.

Model updating provides a hierarchy of responses to such failures. Intercept-only recalibration changes the baseline level of risk while preserving relative predictor effects. Intercept-and-slope recalibration additionally rescales the existing linear predictor. More structural updating re-estimates predictor coefficients or rebuilds the model (Steyerberg et al. 2004). Calibration methods can improve probability estimates without necessarily adding information to the original score (Niculescu-Mizil and Caruana 2005). This hierarchy motivates asking whether the complexity of an adaptation strategy should match the component of the data-generating mechanism that has changed.

For imbalanced binary outcomes, evaluation also benefits from metrics beyond ROC-AUC. Precision-recall summaries emphasize performance on the event class and can reveal behavior that ROC summaries obscure (Davis and Goadrich 2006). Probability-oriented measures such as log loss and the Brier score complement ranking metrics because they respond directly to the quality of the predicted probabilities.

2.3 Distribution shift in credit-risk prediction

Credit-risk prediction has traditionally emphasized discrimination and classification performance, with both statistical and machine-learning approaches widely studied (Bazzana et al. 2024). However, credit portfolios are inherently non-stationary. Sousa, Gama, and Brandão (2016) proposed a dynamic credit-risk framework based on sequential learning from changing populations and showed that static historical models can lose effectiveness as the environment changes. Nikolaidis and Doumpos (2022) studied drift adaptation in behavioral credit scoring using an 11-year proprietary credit-bureau dataset and reported that locally adaptive models can outperform corresponding global models under population drift.

More recent work has made drift itself a central credit-risk modeling problem. Peng and Lessmann (2026) study mortgage default prediction under three concept-drift regimes—sudden, incremental, and recurring—while also manipulating label prevalence, and evaluate both discrimination and calibration in a dynamic survival framework. These studies reinforce the practical importance of non-stationarity, but they primarily focus on developing or comparing adaptive predictive methods.

2.4 Comparison with related studies

The relationship between distribution shift and model updating has been studied previously. Davis and colleagues evaluated changes in outcome prevalence, case mix and predictor–outcome associations,

comparing intercept correction, logistic recalibration and refitting. Their study developed a bootstrap procedure to recommend update complexity, examined different development and updating sample sizes, and included temporal clinical applications (Davis et al. 2019). Thus, the benefits of simple recalibration under some risk-level changes and structural updating under changed predictor effects are established findings.

The present study provides a complementary credit-risk-motivated comparison using specified simulation conditions. An omitted quadratic term makes the interaction between population shift and model approximation error explicit. Density-ratio weighting is compared with recalibration and deployment refitting, while known conditional probabilities permit direct probability-error assessment. Threshold revision is evaluated separately from probability-model updating. These features describe the experiments performed here, rather than establishing that their individual principles are new. The study does not develop an automatic update-selection procedure or validate its descriptive framework on external credit populations.

3 Statistical Framework

3.1 Distribution shift

Development and deployment environments are represented by $P(X, Y)$ and $Q(X, Y)$, respectively (Quiñonero-Candela et al. 2009). Distribution shift occurs when

$$P(X, Y) \neq Q(X, Y).$$

Under covariate shift,

$$P(X) \neq Q(X), \quad P(Y | X) = Q(Y | X).$$

A correctly specified conditional model can remain valid under this change, although finite-sample estimation error may become more consequential in regions poorly represented during development. If the fitted model is misspecified, shifting probability mass toward regions where the approximation is poor can substantially increase predictive error; density-ratio weighting is one established approach for targeting the deployment covariate distribution (Shimodaira 2000).

Outcome-mechanism shift instead satisfies

$$P(Y | X) \neq Q(Y | X).$$

An intercept change in a logistic model changes baseline risk while preserving predictor slopes. A predictor-effect change alters the relationship between at least one predictor and the outcome and can therefore change the ranking induced by the original score.

3.2 Predictive properties

For binary outcomes $Y_i \in \{0, 1\}$ and predicted probabilities $\hat{p}_i$, log loss is

$$-\frac{1}{n} \sum_{i=1}^n [Y_i \log(\hat{p}_i) + (1 - Y_i) \log(1 - \hat{p}_i)].$$

The Brier score is

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{p}_i)^2.$$

In the simulations, probability mean-squared error (PMSE) compares predictions with the known conditional probabilities:

$$\text{PMSE} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - p_i)^2,$$

where p_i is the true conditional event probability specified by the data-generating process. Unlike the Brier score, PMSE does not compare predictions with realized binary outcomes. It is available here because the true probabilities are known in the simulations.

Calibration is assessed through a logistic recalibration model,

$$\text{logit}\{P(Y = 1)\} = a + b \text{logit}(\hat{p}),$$

where $a = 0$ and $b = 1$ represent ideal calibration (Van Calster et al. 2019). ROC-AUC measures discrimination rather than numerical probability accuracy (Steyerberg et al. 2010).

Threshold-based performance is treated separately. Balanced accuracy is

$$\text{BA} = \frac{1}{2} (\text{Sensitivity} + \text{Specificity}).$$

The probability model and the operating threshold are therefore distinct components of the predictive system.

4 Empirical Loan-Default Application

4.1 Data and development design

The analysis uses the *Loan Default Prediction Dataset* shared on Kaggle by NIKHIL (NIKHIL 2023). It contains 255,347 records, including 29,653 defaults (11.6%). Sixteen borrower and loan characteristics, together with a loan-to-annual-income ratio, are used as predictors. LoanID is excluded from model fitting. The source does not establish the original collection process, observation horizon or whether the records are actual or synthetic, so the data are treated as a prediction benchmark.

The data are divided into four non-overlapping partitions: 60% for training, 15% for calibration, 10% for validation, and 15% for final testing. Preprocessing is estimated using training data only. Logistic regression, ridge logistic regression, random forest, and gradient boosting are compared using validation log loss as the primary probability-quality criterion. Models within 0.001 log-loss units of the best candidate are considered practically tied; the code prefers the lower complexity rank, followed by lower log loss and Brier score. The empirical split uses seed 42, whereas the controlled simulations use seed 1998. This comparison reflects the continued use of both classical statistical and machine-learning classifiers in default prediction (Bazzana et al. 2024).

4.2 Internal predictive performance

Raw logistic regression achieved the lowest validation log loss, although its performance was close to Platt-calibrated logistic regression. On the locked test set, the selected model achieved ROC-AUC 0.7585, PR-AUC 0.3288, Brier score 0.0908, and log loss 0.3111. ROC-AUC and PR-AUC are reported together because the event rate is modest and the two summaries emphasize different aspects of binary ranking performance (Davis and Goadrich 2006).

A validation-selected threshold of 0.1175 maximized balanced accuracy. On the test set, this threshold produced recall 0.6843 and specificity 0.7011. Test-set calibration was close to ideal, with calibration intercept approximately 0.0073 and slope approximately 0.9989.

These results establish internal predictive performance. They do not establish stability under a different deployment distribution.

5 Monte Carlo Design

5.1 Baseline data-generating process

The simulation uses three independent standard-normal predictors,

$$X_1, X_2, X_3 \stackrel{iid}{\sim} N(0, 1),$$

with baseline conditional outcome model

$$\text{logit}(p) = \beta_0 + 0.8X_1 - 0.6X_2 + 0.4X_3.$$

The intercept is chosen to yield an event prevalence near 12%. Each condition uses 500 Monte Carlo replications with seed 1998. Unless otherwise stated, the development training sample, deployment adaptation sample, and deployment test sample each contain 5,000 observations. Fresh independent samples are generated in every replication so that adaptation is never evaluated on the same observations used to estimate the adaptation parameters.

Because the data-generating process is known, the true conditional probability is available as an oracle benchmark. This permits separation of intrinsic prediction difficulty from error caused by estimation, misspecification, or failure to adapt.

5.2 Experiment 1: covariate shift and misspecification

The true model includes an additional quadratic term,

$$\text{logit}(p) = \beta_0 + 0.8X_1 - 0.6X_2 + 0.4X_3 + \beta_4X_1^2,$$

while the fitted development model omits X_1^2 . Misspecification is varied through β_4 , and deployment shift is introduced through

$$X_1 \sim N(\delta, 1).$$

The misspecification parameter is varied over $\beta_4 \in \{0, 0.25, 0.50\}$, and the covariate-shift parameter over $\delta \in \{0, 0.5, 1, 1.5, 2\}$. For each value of β_4 , the intercept β_0 is calibrated at $\delta = 0$ so that baseline

event prevalence remains approximately 12%. The fitted logistic model deliberately omits X_1^2 , making $\beta_4 = 0$ the correctly specified control and the positive values progressively stronger misspecification. The conditional outcome mechanism is unchanged between development and deployment.

5.3 Experiment 2: intercept outcome shift

The development model is

$$\text{logit}(p_P) = \beta_0 + 0.8X_1 - 0.6X_2 + 0.4X_3,$$

while deployment follows

$$\text{logit}(p_Q) = \beta_0 + \gamma + 0.8X_1 - 0.6X_2 + 0.4X_3.$$

The intercept-shift parameter is varied over $\gamma \in \{0, 0.25, 0.50, 0.75, 1\}$. Thus baseline risk changes while predictor effects remain fixed.

5.4 Experiment 3: predictor-effect shift

The effect of X_1 changes according to

$$\beta_{1,\text{deploy}} = 0.8 - \kappa,$$

with $\kappa \in \{0, 0.4, 0.8, 1.2, 1.6\}$. The deployment intercept is recalibrated so that prevalence remains near 12%, isolating structural change in the predictor–outcome relationship from prevalence change.

6 Failure Patterns Under Distribution Shift

The controlled experiments show that different shift mechanisms damage different parts of the predictive system. Table 1 summarizes the strongest condition from each experiment before adaptation. Table 1 reports the retained summaries from the reconstructed initial failure-pattern experiments, whereas Table 2 and Figures 1–3 report the separately executed adaptation experiments. Their unadapted estimates therefore differ slightly because they use different simulated samples. These rows should be interpreted as mechanism contrasts rather than as directly comparable real-world scenarios because the three experiments perturb different components of the data-generating process.

Table 1: Predictive behavior at the strongest pre-adaptation shift condition. PMSE measures squared error against the known conditional probabilities.

Mechanism	Condition	Event rate	PMSE	ROC-AUC	Calibration intercept	Calibration slope
Covariate shift + misspecification	$\beta_4 = 0.50$ $\delta = 2.0$	0.5908	0.0444	0.8881	1.221	1.815
Intercept outcome shift	$\gamma = 1.0$	0.2435	0.0200	0.7562	0.998	0.998
Predictor-effect shift	$\kappa = 1.6$	0.1201	0.0269	0.4726	-2.213	-0.090

Under correctly specified covariate shift, fitted probabilities remain close to the oracle. When nonlinear structure is omitted, however, the same shift increasingly exposes misspecification. Under strong misspecification, probability mean-squared error rises sharply as the deployment distribution moves away from the development population. Discrimination can simultaneously improve because the shifted population becomes more separable, illustrating that increasing ROC-AUC does not imply improving probability accuracy.

Under intercept outcome shift, the calibration intercept tracks the imposed shift while the calibration slope remains near one. Probability error and log loss deteriorate, but the fitted model remains close to the oracle in ranking. This is the characteristic pattern of a baseline-risk change and is consistent with the distinction between discrimination and calibration emphasized in prediction-model evaluation (Steyerberg et al. 2010; Van Calster et al. 2019).

Under predictor-effect shift, the original model loses discrimination as the effect of X_1 weakens and reverses. At the strongest shift, the original model’s ROC-AUC falls from approximately 0.762 to 0.473 while the oracle remains informative. This represents structural obsolescence rather than simple probability-scale drift.

Decision thresholds form a separate source of instability. Under covariate shift, the balanced-accuracy-optimal threshold moves substantially even when the conditional probability model is correctly specified. Model validity and decision-rule validity are therefore not equivalent.

7 Adaptation After Distribution Shift

7.1 Adaptation strategies

The adaptation experiment compares interventions that modify different components of the predictive system. The development model is first fitted on a baseline training sample of $n = 5000$. For recalibration and deployment refitting, adaptation parameters are estimated on an independent labeled deployment sample of $n = 5000$, and all reported performance is then computed on a separate deployment test sample of $n = 5000$. In the covariate-shift adaptation experiment, the misspecification level is fixed at the strong condition $\beta_4 = 0.50$, while $\delta \in \{0, 0.5, 1, 1.5, 2\}$.

Let $\hat{\eta}_{\text{old}} = \text{logit}(\hat{p}_{\text{old}})$ denote the linear predictor from the frozen development model.

No adaptation. The original probabilities $\hat{p}_{\text{old}}$ are used without modification.

Intercept recalibration. On the deployment adaptation sample, a logistic model with $\hat{\eta}_{\text{old}}$ as an offset estimates only a new intercept a :

$$\text{logit}\{P(Y = 1)\} = a + \hat{\eta}_{\text{old}}.$$

This intervention changes baseline risk while preserving the original score scale and ranking.

Intercept-and-slope recalibration. A logistic model is fitted on the adaptation sample using the old linear predictor as the sole covariate:

$$\text{logit}\{P(Y = 1)\} = a + b\hat{\eta}_{\text{old}}.$$

This can shift, rescale, or reverse the existing one-dimensional score but cannot separately change the coefficients of its component predictors. These recalibration strategies correspond to established model-updating approaches (Steyerberg et al. 2004).

Importance-weighted refitting. This intervention is used only in the covariate-shift experiment. Because baseline $X_1 \sim N(0, 1)$ and deployment $X_1 \sim N(\delta, 1)$, the exact density ratio is

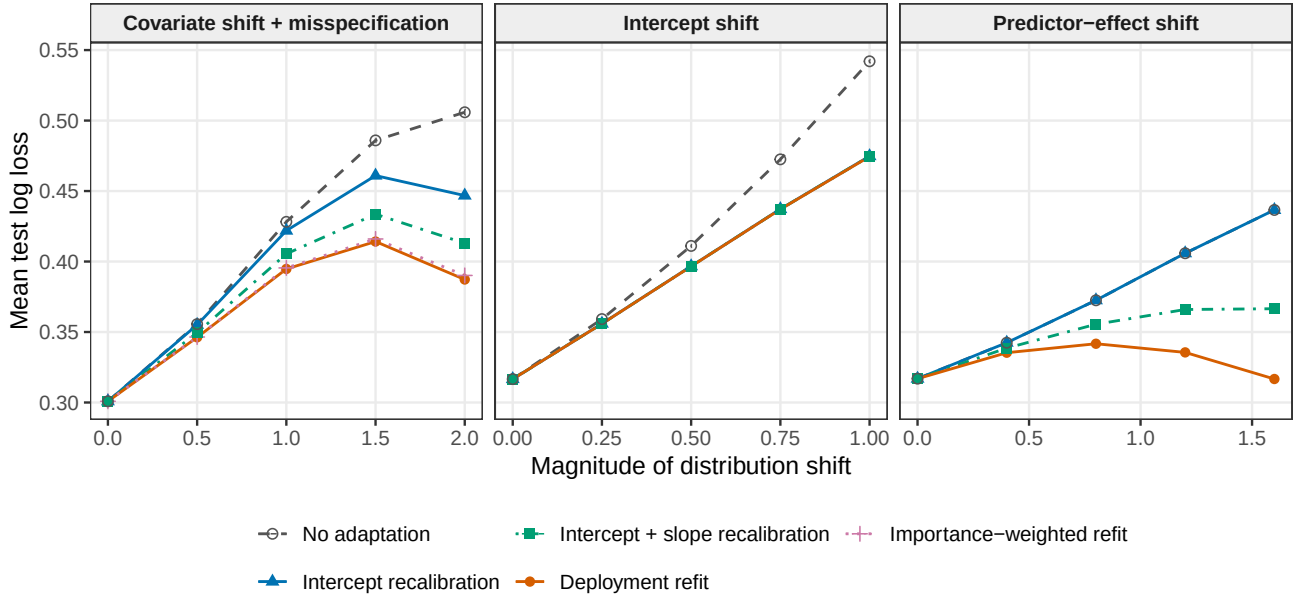
$$w(X_1) = \frac{q(X_1)}{p(X_1)} = \exp\left(\delta X_1 - \frac{\delta^2}{2}\right).$$

The density-ratio weights are evaluated on the baseline training observations. To reduce the influence of extreme finite-sample weights, the weights are capped at their empirical 99th percentile within each training sample; the capped weights are then normalized to have mean one. The resulting weights are used in a weighted logistic refit of the same linear model class. Thus the method changes the target covariate distribution represented by the fitting criterion but does not add the omitted quadratic term. This implements the density-ratio logic of covariate-shift weighting (Shimodaira 2000).

Deployment refitting. A new logistic model is estimated directly from the labeled deployment adaptation sample using the same predictor set available to the fitted model. Under predictor-effect shift this allows the individual predictor coefficients to change; under the misspecified covariate-shift experiment it remains a linear refit and therefore does not receive knowledge of the omitted quadratic term.

Threshold adaptation. Threshold revision is evaluated separately because it changes the operating rule rather than the probability model. A baseline threshold maximizing balanced accuracy is estimated from an independent baseline validation sample. For each shifted condition, a new threshold is selected on an independent shifted tuning sample and compared with the frozen baseline threshold on a separate shifted test sample. This separation avoids evaluating a threshold on the same observations used to select it.

7.2 Probability performance



Bars: approximate 95% Monte Carlo intervals; 500 replications per condition.

Figure 1: Probability performance after distribution shift. Lower log loss indicates more accurate probabilistic predictions.

Table 2: Adaptation performance at the strongest shift condition from each experiment. Lower log loss and Brier score are better; higher ROC-AUC indicates stronger discrimination.

Shift	Method	Log loss	Brier	ROC-AUC
Covariate + misspecification	Deployment refit	0.3873	0.1238	0.9039
Covariate + misspecification	Importance-weighted refit	0.3901	0.1247	0.9031
Covariate + misspecification	Intercept recalibration	0.4468	0.1428	0.8881
Covariate + misspecification	Intercept + slope recalibration	0.4130	0.1335	0.8881
Covariate + misspecification	No adaptation	0.5059	0.1676	0.8881
Intercept shift	Deployment refit	0.4745	0.1544	0.7556
Intercept shift	Intercept recalibration	0.4747	0.1545	0.7554
Intercept shift	Intercept + slope recalibration	0.4747	0.1544	0.7554
Intercept shift	No adaptation	0.5420	0.1743	0.7554
Predictor-effect shift	Deployment refit	0.3167	0.0933	0.7627
Predictor-effect shift	Intercept recalibration	0.4366	0.1202	0.4719
Predictor-effect shift	Intercept + slope recalibration	0.3665	0.1055	0.5260
Predictor-effect shift	No adaptation	0.4366	0.1201	0.4719

At the strongest covariate-shift condition with misspecification, mean log loss is 0.5059 without adaptation, 0.4468 after intercept recalibration, 0.4130 after intercept-and-slope recalibration, 0.3901 after importance-weighted refitting, and 0.3873 after deployment refitting. The ordering shows that global score recalibration repairs part of the probability error but does not remove the consequences of omitted nonlinear structure.

For the strongest intercept shift ($\gamma = 1$), mean log loss decreases from 0.5420 without adaptation to approximately 0.4747 after intercept recalibration, essentially matching deployment refitting at 0.4745. This is consistent with the data-generating mechanism: the shift is primarily an intercept change, so an intercept correction directly targets the changed component.

For the strongest predictor-effect shift ($\kappa = 1.6$), mean log loss is 0.4366 without adaptation, 0.3665 after intercept-and-slope recalibration, and 0.3167 after deployment refitting. Recalibration therefore improves probability performance but does not fully reconstruct the changed conditional model.

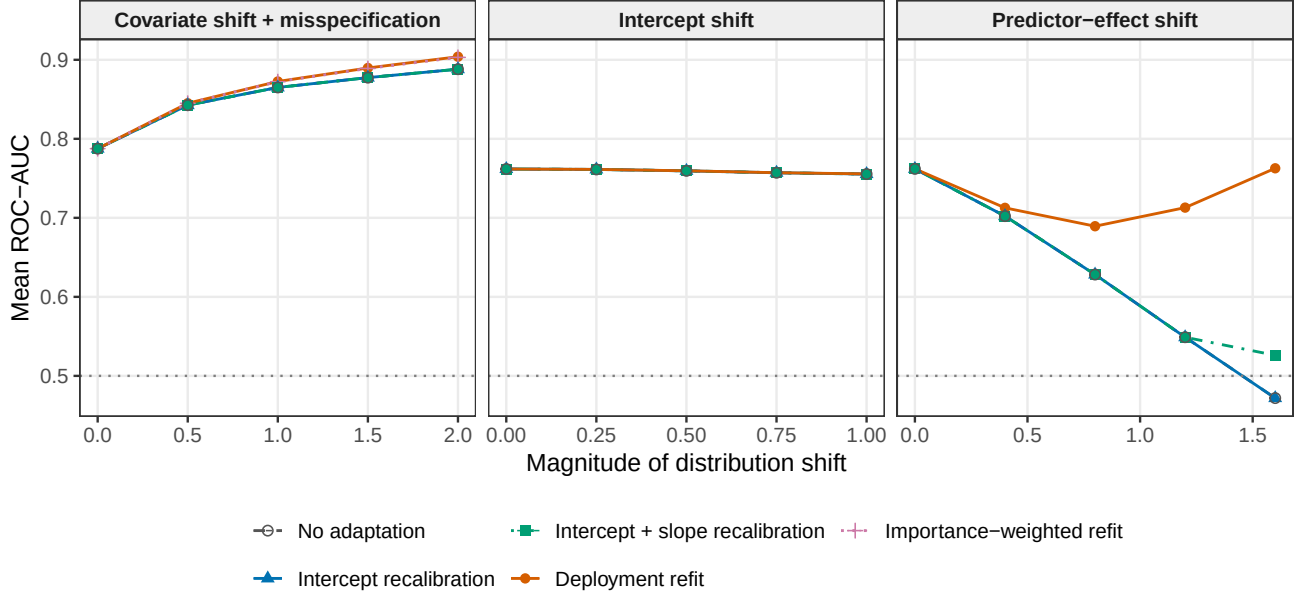
7.3 Discrimination

Recalibration has little effect on discrimination under intercept shift because the ordering induced by the original score remains useful. The strongest contrast occurs under predictor-effect shift. At $\kappa = 1.6$, the unadapted model has mean ROC-AUC 0.4719. Intercept-and-slope recalibration raises mean ROC-AUC to 0.5260, while deployment refitting reaches 0.7627.

The partial AUC recovery from intercept-and-slope recalibration is possible because the fitted recalibration slope can reverse the direction of the old score. It cannot, however, separately revise the coefficients of the individual predictors. Deployment refitting can estimate the changed multivariable relationship and therefore recovers substantially more discrimination.

7.4 Calibration instability under severe structural change

At the strongest predictor-effect shift ($\kappa = 1.6$), test-sample calibration coefficients after intercept-and-slope recalibration varied widely across the 500 replications. The mean intercept and slope were



Bars: approximate 95% Monte Carlo intervals; 500 replications per condition.

Figure 2: Discrimination after distribution shift and adaptation. Score recalibration does not fully recover discrimination lost after structural change.

-38.24 and -17.97, whereas their medians were -0.049 and 0.969. A diagnostic rerun reproduced the original adaptation results and showed that the extreme coefficients occurred when the adaptation recalibration slope was close to zero, leaving very little variation in the updated test scores.

For example, replication 251 had an adaptation slope of 0.00001586 and a standard deviation of 0.00001690 for the updated test log-odds scores. The subsequent test-calibration fit converged but produced an intercept of -19,243.45 and a slope of -9,547.42. Across this condition, 13 of 500 replications had absolute test-calibration slopes greater than 10; all 13 had absolute adaptation slopes below 0.01515. All recorded test-calibration fits converged, so convergence alone did not establish stable calibration estimates.

These results show why the arithmetic means should not be interpreted as typical calibration performance. The near-ideal medians also conceal substantial variation and do not establish satisfactory overall performance. Calibration coefficients should therefore be interpreted alongside their distributions, probability loss, Brier score, and discrimination. All replications were retained.

7.5 Decision-threshold adaptation

Threshold adaptation produces its largest benefit under covariate shift with misspecification. At the strongest condition, the mean baseline threshold is 0.1598 and the adapted threshold is 0.3940. Mean balanced accuracy rises from 0.6555 to 0.8031, a gain of 0.1475 with Monte Carlo standard error approximately 0.00113.

Under the strongest intercept shift, threshold adaptation yields only a small balanced-accuracy gain of approximately 0.0025. Under the strongest predictor-effect shift, the gain is approximately 0.0120, while balanced accuracy remains near 0.494. A threshold can change the operating point of an existing score, but it cannot recover ranking information that the score no longer contains.

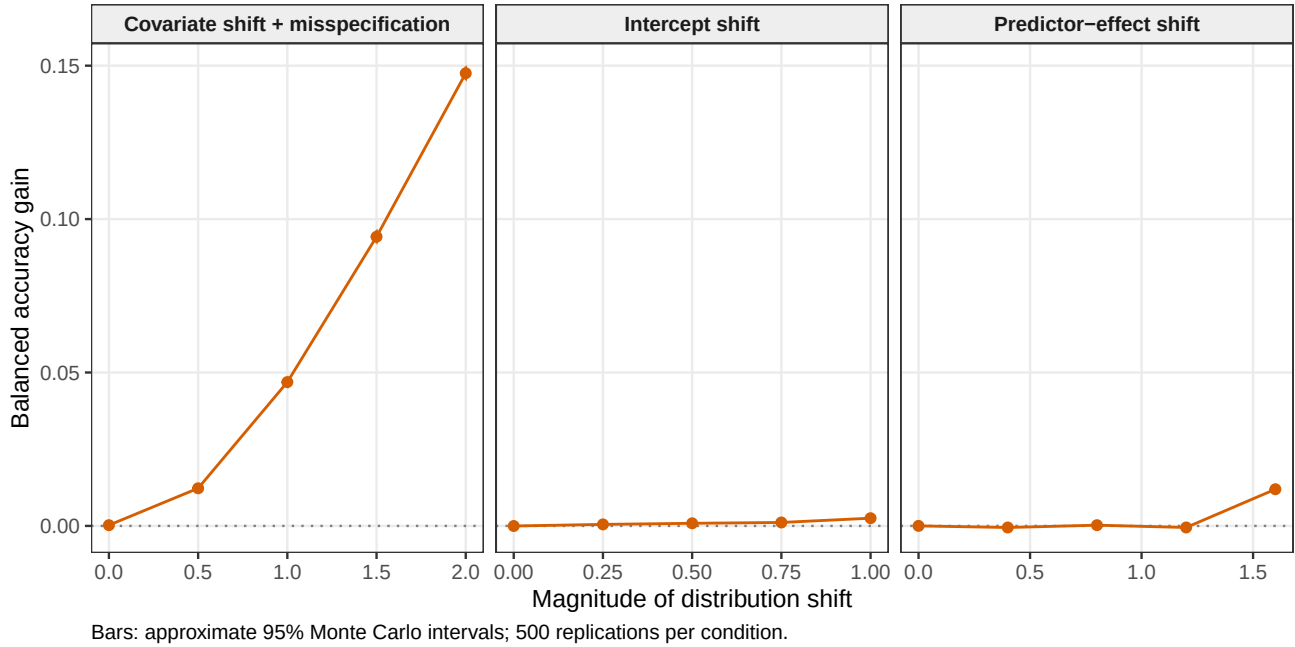


Figure 3: Benefit of adapting the decision threshold. Points show gain in balanced accuracy relative to retaining the baseline threshold.

8 Failure–Response Framework

The simulation results suggest a mechanism-based interpretation of model maintenance. The framework is descriptive rather than a universal diagnostic rule: similar empirical symptoms can arise from multiple underlying causes, and observed deployment data should be investigated directly.

Table 3: Failure–response framework linking observed predictive patterns to plausible statistical changes and adaptation responses.

Observed pattern	Plausible statistical change	Response and evidence basis
Calibration intercept drifts while ranking remains stable	Baseline-risk/intercept shift	Intercept recalibration
Probability scale is distorted but score remains informative	Possible calibration-scale change; not isolated as a separate shift experiment here	Intercept-and-slope recalibration; established updating rationale, with partial repair observed in the present experiments
Predictor population changes and misspecification becomes consequential	Change in $P(X)$ interacting with model approximation error	Reweighting, recalibration, and possible refitting/redevelopment
Discrimination deteriorates because predictor effects change	Structural change in $P(Y X)$	Deployment refitting or model redevelopment
Probability model remains useful but operating performance changes	Decision environment/population change	Threshold reassessment

The central implication is

probability-model repair $\neq$ ranking repair $\neq$ decision-rule repair.

9 Discussion

The experiments demonstrate that deployment stability cannot be summarized by a single performance measure. Covariate shift, baseline-risk shift, and predictor-effect shift generate different combinations of calibration error, probability loss, discrimination change, and threshold instability.

The intercept-shift experiment provides the cleanest example of targeted repair and is consistent with the model-updating principle that parsimonious recalibration can be preferable when the original score structure remains useful (Steyerberg et al. 2004). Because the deployment mechanism differs from development primarily through baseline risk, intercept recalibration nearly matches full deployment refitting. Re-estimating the entire model is unnecessary in this controlled setting. The separate supplementary sample-size experiment further shows that intercept recalibration has lower mean log loss than full deployment refitting with 250, 500 and 1,000 adaptation observations. When the changed mechanism can be addressed by a single intercept adjustment, estimating additional coefficients introduces avoidable estimation variability. Under predictor-effect shift and covariate shift with misspecification, deployment refitting instead has the lowest mean log loss among the methods tested at all four adaptation sizes. These comparisons concern the specified supporting design and are not universal sample-size recommendations.

Covariate shift with misspecification produces a different pattern. The conditional mechanism is unchanged, but the development model is an imperfect approximation. As deployment moves toward regions where the omitted nonlinear component matters more, simple recalibration cannot recover all lost probability accuracy. Importance weighting and deployment refitting perform substantially better.

Predictor-effect shift is more severe because the original risk score itself becomes structurally obsolete. Score recalibration can rescale or reverse that one-dimensional score but cannot separately modify its component predictor effects. Consequently, it partially improves performance but remains far below deployment refitting in discrimination.

The threshold experiments reinforce the distinction between statistical prediction and operational decision rules. A model may remain probabilistically meaningful while the threshold that was optimal in the development population becomes unsuitable after population change. Conversely, changing a threshold cannot repair a score whose ranking information has deteriorated.

For model monitoring, this implies that ROC-AUC alone is insufficient; discrimination and calibration should be evaluated as distinct properties (Steyerberg et al. 2010; Van Calster et al. 2019). Deployment assessment should consider changes in the predictor distribution, event prevalence, calibration, discrimination, probability loss where outcomes become available, and the performance of the operating decision rule.

10 Limitations and Future Research

10.1 Limitations

The study has several limitations. First, the Monte Carlo experiments are intentionally stylized. They isolate mechanisms rather than reproduce every feature of real credit portfolios. Second, the empirical analysis uses one publicly available loan dataset and does not provide genuine temporal or external

validation. Third, the recorded default indicator does not have a documented observation horizon, so empirical estimates are interpreted as probabilities of the recorded outcome rather than formal fixed-horizon probabilities of default.

Fourth, the adaptation experiments assume access to labeled deployment observations. The amount and timing of labeled data available in practice may materially affect adaptation performance. The supplementary experiment varies the number of labels but does not examine delayed outcomes, selective labeling, or dependence between observations. Its sample-size comparison excludes importance weighting and threshold revision. The supplementary nonlinear mechanism also uses a different intercept setup from the original experiment, so its numerical results should not be pooled with the original results. Fifth, balanced accuracy is used as a statistical threshold objective because verified monetary costs of false positives and false negatives are unavailable. The selected threshold should therefore not be interpreted as economically optimal.

Finally, the failure-response framework is mechanistic rather than diagnostic. Observing calibration drift or declining AUC in practice does not uniquely identify the data-generating change responsible for it.

10.2 Future research

Several extensions follow directly from the present design. The supplementary sample-size experiment examines one aspect of limited deployment labels; broader sample-size designs and regularized updates remain to be investigated. Correlated and heavier-tailed predictors could be introduced to assess robustness beyond the independent Gaussian design. Temporal or external credit data would allow the framework to be evaluated under naturally occurring rather than synthetic distribution shift.

Further work could also replace balanced accuracy with explicit asymmetric decision costs, study uncertainty in threshold selection, and develop formal procedures for deciding when a deployed model should be recalibrated, refitted, or replaced.

11 Reproducibility and Data/Code Availability

11.1 Reproducibility

The empirical pipeline was rerun by the author in RStudio with forced retraining, seed 42 and neural-network fitting disabled. Computed validation and test tables in the resulting PDF matched the publication manuscript’s principal empirical values at displayed precision. This check compares the rendered results; an independent audit of the saved empirical model objects and input files remains outstanding.

The controlled simulations used seed 1998 and 500 replications per condition. The initial script was reconstructed because its original standalone source was not preserved. Table 1 uses retained rerun summaries recalculated from replication-level records. The threshold-stability comparison passed the specified tolerances against the available frozen reference; corresponding reference files for coefficient recovery, intercept shift and predictor-effect shift were unavailable. This establishes partial numerical agreement, not exact reproduction of the original analysis.

Adaptation summaries were independently recalculated from the supplied raw records. An instrumented rerun supported the calibration-instability explanation in Section 7.4. The two separate supporting experiments used 1,000 replications each. All 399 supporting adaptation-summary checks and 512 sensitivity mean/Monte Carlo standard-error checks agreed within an absolute tolerance of 10^{-11} . Replication keys were unique, numerical records were finite, and sensitivity success/failure counts

matched. Shared test-sample quantities and unchanged-model metrics were consistent across methods and sample sizes. Supporting scripts and settings were supplied, including the recovered adaptation script. These checks verify summaries of replication-level metrics; they do not recompute borrower-level predictions or independently verify execution. Scripts, settings, results and session information are retained locally. The supporting experiments are neither external validation nor exact replications of the original design.

11.2 Data availability

The benchmark dataset is publicly available from <https://www.kaggle.com/datasets/nikhil9/loan-default> (NIKHIL 2023). The source advertises a CC0: Public Domain license and attributes the dataset to Coursera’s Loan Default Prediction Challenge. The original collection process and default observation horizon remain undocumented. A comparison of the supplied CSV with the Kaggle download yielded the same SHA-256 checksum: `1d7556a9071e7f9e872dc05a0cad174229fb1164b1cf3470ad35eed195c24278`. Input acquisition instructions and fingerprints are provided in the project repository at <https://github.com/sdbambill/predictive-stability-distribution-shift>. The benchmark CSV and synthetic stress-test input are not redistributed in that repository. The synthetic stress-test dataset is not an external validation dataset.

The repository includes retained failure-pattern summaries, adaptation and threshold summaries, supporting sample-size summaries, and selected verification records. Some complete replication-level records remain retained locally by the author. The deposited summaries support the manuscript tables and figures; their agreement with recorded replication-level metrics does not constitute independent re-execution of the underlying models.

11.3 Code availability

The publication and supplementary R Markdown sources, bibliography, supporting empirical pipeline, reconstructed initial simulation script, recovered 500-replication adaptation script, rendering scripts, and selected verification records are available at <https://github.com/sdbambill/predictive-stability-distribution-shift>. The initial simulation script reconstructs the documented design; it is not the preserved original standalone source. The repository does not yet contain every script, setting, and full replication-level record for the separate supporting experiments or instrumented calibration diagnostic. Recorded R session information is supplied, but an original environment lockfile has not yet been deposited.

The repository documents the scope and limitations of the file-consistency checks. Reproducibility checks and their limits are described in Section 11.1. A permanent Zenodo archive and DOI are pending.

11.4 Use of generative AI

ChatGPT was used to assist with R code generation or reconstruction and manuscript drafting. The author developed the research idea, conducted the analyses in RStudio, reviewed the implementation against the intended mathematics and assumptions, and takes responsibility for the manuscript and analysis.

12 Conclusion

Different forms of distribution shift produce different predictive failure modes. Baseline-risk shift can substantially damage probability calibration while preserving ranking and can be repaired efficiently by intercept recalibration. Covariate shift can expose model misspecification, in which case broader adaptation or refitting may be required. Changes in predictor effects can invalidate the original ranking itself, making score recalibration insufficient and favoring model redevelopment.

The supplementary sample-size results show that the amount of labeled deployment data also matters: simple intercept recalibration performs well with limited data when baseline risk alone changes, whereas broader refitting is required to address changed predictor relationships in the tested conditions. Updating an unchanged model with small samples can worsen average performance.

The broader lesson is that internal predictive performance is not evidence of deployment stability. Calibration, discrimination, probability accuracy, and decision-threshold performance are distinct properties, and the appropriate corrective response depends on which property has become unstable.

The supporting experimental design, detailed results, figure and table are provided in the separate supplementary document.

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